

# Final Project Report

**Course Code and Name:** BIOI602/DATA602/MSML602-PCS2: Principles of Data Science

**Semester and Year:** Fall, 2024

**Instructor Name:** Dr. Fardina Fathmiul Alam

**Group Project Name:** Breast Tumor Prediction using Random Forest, PCA and SVM

**URL to Final Tutorial:** <https://rohithreddymandla.github.io/>

**Group Members:**

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## Contributions:

| Group Member            | Contributions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rohith Reddy Mandla     | <p><b>Project Idea:</b> Conceptualized the main issue, explored the relevant models and the corresponding questions to be answered through the analysis.</p> <p><b>Dataset Selection and Preprocessing:</b> Worked on gathering datasets, analysis and pre-processing of data.</p> <p><b>Collaboration on other Tasks:</b> Also helped with tasks related to ML Design, Data Exploration and final report creation.</p>                                                                               |
| Manasanjani Kondapi     | <p><b>ML Algorithm Design/Development:</b> Worked on developing the model, feature selection, designed the code to implement the algorithms</p> <p><b>Data Exploration and Summary Statistics:</b> Also performed data cleaning, conducted exploratory data analysis, summarized key statistics and analyzed the results.</p> <p><b>Collaboration on other Tasks:</b> Also helped with the other tasks related to data pre-processing and visualization.</p>                                          |
| Krishnendra Singh Tomar | <p><b>Visualization, Result Analysis, Conclusion:</b> Created important visualizations for analysing and interpreting the results. Also worked on deriving the final conclusions from the results.</p> <p><b>Final Tutorial Report Creation:</b> In addition to the analysis, documented the project tutorial which include code, visualizations, and precise writing.</p> <p><b>Collaboration on other Tasks:</b> Also helped with tasks related to questions to consider, and data exploration.</p> |

# 1 Introduction:

## Topic Motivation:

It is irrefutable fact that Breast Cancer is one of the main causes of cancer-related deaths worldwide. So, it is crucial for a variety of stakeholders (including healthcare professionals, patients, and medical researchers) to address the problem of breast tumor detection and significantly improve the diagnostic process. An accurate, reliable and efficient model allows early detection and timely intervention, leading to higher survival rates and reduced treatment costs due to the ability to catch tumors in their early stages when they are easier to treat. Patients will benefit from more accurate diagnoses and personalized treatment plans, leading to better outcomes and quality of life. As part of this project, we mainly focused on designing and developing a robust machine learning model which can accurately predict the breast tumor presence. In order to achieve this, we listed the main questions, like the dataset selection, optimization techniques, model selection, important metrics to determine the performance.

## Importance of Answering These Questions:

A proper dataset, that includes relevant features and valuable, accurate information is utmost important, to train the best machine learning model. So, we used Breast Cancer Wisconsin (Diagnostic) Dataset for this project. Not only dataset, but also selection and optimization of the machine learning models is essential. Hence, we selected Random Forest and SVM models for this use case. Also, the PCA technique is used for dimensionality reduction and to minimize complexity as well as the training time. By this, we aim to achieve the optimal balance between performance and computational efficiency to ensure the resulting system can be deployed in real-world clinical environments, where both accuracy and speed are essential. We also ensured, using these models and techniques, the performance metrics such as precision, recall, F1-score are within the acceptable limits. These metrics are critical, as they provide a comprehensive evaluation of the model's ability to correctly identify both positive and negative cases, minimizing false positives and negatives, which is vital for medical applications. Machine learning models can vary significantly in terms of efficiency, and it is essential to select the most appropriate model for tumor detection. In addition, the ability to replicate the results of this model is crucial to building trust in the performance of the system and to ensure that it can be widely adopted in different healthcare institutions.

# 2 Data Curation:

## Datasets Chosen:

### Breast Cancer Wisconsin (Diagnostic) Dataset

W. Wolberg, O. Mangasarian, N. Street, and W. Street. "Breast Cancer Wisconsin (Diagnostic)," UCI Machine Learning Repository, 1993. [Online]. Available: <https://doi.org/10.24432/C5DW2B>.

**Source(s) of Data:** UCI Machine Learning Repository

## Data Description:

- **Relevance:**

This dataset has 569 instances and 30 features, which are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. It is also multivariate and hence, is relevant and informative, to address our problem and provides accurate data for the project's objective.

- **Feature Set:**

The dataset includes real-world features like radius, texture, and perimeter, etc., which are useful for statistical analysis and to develop an effective classifier model.

- **Reliability:**

This dataset is widely cited and is licensed under CC BY 4.0, thus assuring the credibility and reliability.

- **Advanced Techniques:**

To develop an accurate and effective model, advanced techniques like PCA, SVM, Random Forest are required. This dataset enables incorporation of such methods with ease.

**Data Transformation:** We referred to the UCI Machine Learning Repository to collect the original, raw dataset. It is then cleaned and pre-processed by handling the missing values, by normalizing the continuous features and converting categorical data to the numerical features. This formatting allows for quick and easy access to the rows and columns, thus supporting direct use in machine learning algorithms. The cleaned data is organized into a **Pandas DataFrame** for effective data manipulation and exploratory analysis.

### 3 Exploratory Data Analysis:

**Data Cleaning:** Missing Values, outliers, etc are then removed from the dataset and consistency is ensured. Imputation methods like mean(for continuous values) and mode(for categorical values) are used to handle the missing values. IQR method was incorporated to detect and handle the outliers.

**T-test Results:** T-test is performed, as part of Hypothesis testing to compare means of continuous features for both the benign and malignant tumor categories. The null hypothesis ( $H_0$ ) assumes means are same, whereas, the alternative ( $H_a$ ) suggests a significant variation between them. Features like 'mean\_Area', 'mean\_Concave\_pts', and 'mean\_Radius' resulted in p-values below 0.05, thus leading to the rejection of the null hypothesis and this indicates their significance in distinguishing tumor types.

**ANOVA Results:** Analysis of Variance (ANOVA) was conducted to compare means across more than two groups. The null hypothesis ( $H_0$ ) assumes means are same, while the alternative ( $H_a$ ) suggests at least one mean is different from others. This is confirmed by ANOVA, by which, features like 'worst\_Concave\_pts', 'mean\_Area', and 'mean\_Perimeter' showed significant differences between tumor categories, with p-values less than 0.05.

**Chi-squared Test Results:** The Chi-squared test resulted in p-values significantly smaller than 0.05. This strongly support our alternate hypothesis( $H_a$ ) which suggests important association between categorical features (e.g., tumor type and smoothness classification).

**Interpretation of Results:** The p-value threshold of 0.05 was used for statistical significance in both T-tests and ANOVA. It is clear that, features such as 'mean\_Concave\_pts' and 'mean\_Radius' showed p-values below 0.05, indicating their significance in distinguishing between benign and malignant tumor categories. Whereas, few other features, like 'fractal\_dimension\_error', did not differ much.

**Visualization:** Post the T-Test, a bar plot was used to visualize the exact correlations of features with tumor classification, highlighting relevant features like 'mean\_Concave\_pts', 'worst\_Concave\_pts', and 'mean\_Radius'. A box plot after ANOVA is used to depict how features like 'worst\_Perimeter' and 'mean\_Area' distinctly separate the tumor types.

**Correlation Matrix:** After the Chi-squared test, a correlation matrix was calculated to analyse relationships among continuous features which presented strong correlations between 'mean\_Perimeter', 'mean\_Area', and 'worst\_Perimeter', which will be addressed in future feature selection steps to mitigate multicollinearity.

## 4 Primary/Machine Learning Analysis:

### Machine Learning Techniques:

We employed two techniques to evaluate tumor classification:

1. **Univariate Feature Selection with Random Forest** and
2. **PCA with Support Vector Machines (SVM)**. These were chosen to assess their ability to distinguish benign and malignant tumors using the selected features.

#### 1. Feature Selection and Random Forest:

Univariate Feature Selection identified the top five features:

$$mean\_Perimeter, \quad mean\_Area, \quad se\_Area, \quad worst\_Perimeter, \quad worst\_Area$$

Random Forest, an ensemble method, achieved an accuracy of 96% by aggregating predictions from multiple decision trees. The prediction is averaged over  $T$  trees:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T f_t(x)$$

#### Code for Random Forest:

```
# Train Test Split
from sklearn.model_selection import train_test_split
y = df['Target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.33, random_state=42)
# Training the Random Forest Classifier
from sklearn.ensemble import RandomForestClassifier
rfc = RandomForestClassifier(n_estimators=200)
rfc.fit(X_train, y_train)
y_pred = rfc.predict(X_test)
```

#### 2. Principal Component Analysis (PCA) and SVM:

PCA reduces dimensionality by projecting the data onto new components. These transformed features are then classified using SVM, which separates classes with an optimal hyperplane. This method achieved 97% accuracy. The covariance matrix  $\Sigma$  is computed as:

$$\Sigma = \frac{1}{n-1} X^T X$$

and the principal components  $W$  are found by solving:

$$\Sigma W = \lambda W$$

The transformed data is projected as:

$$X_{new} = X \cdot W$$

This approach works effectively with SVM for classification.

### Code for PCA:

```
# Feature Scaling
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
scaler.fit(df_features)
features_scaled = scaler.transform(df_features)
X_scaled = pd.DataFrame(data=features_scaled, columns=df_features.
                        columns)

# PCA Transformation
from sklearn.decomposition import PCA
pca = PCA(n_components=2)
pca.fit(X_scaled)
X_pca = pca.transform(X_scaled)
```

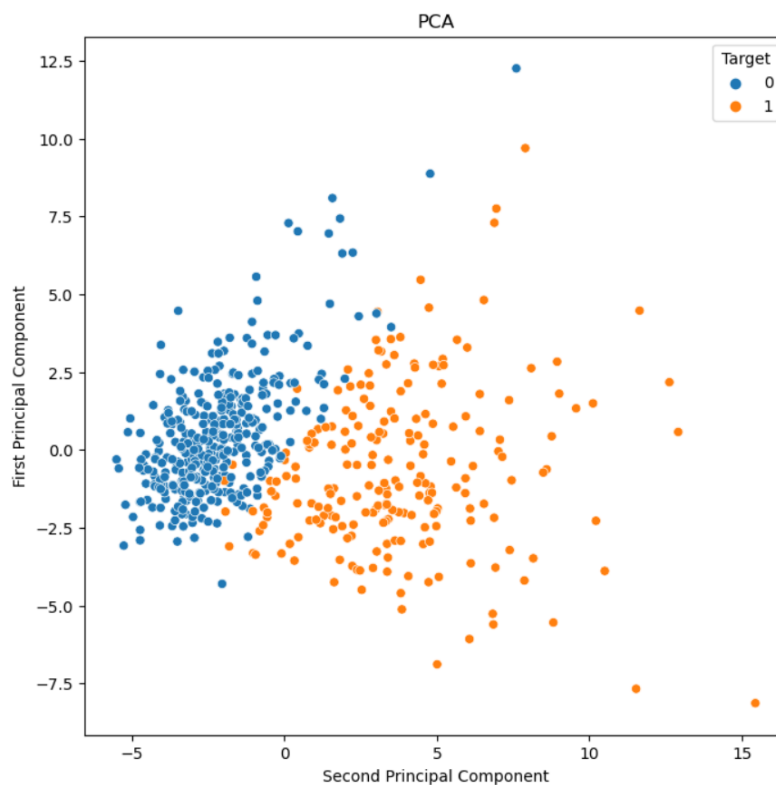


Figure 1: PCA Visualization

### Code for SVM:

```
# Train Test Split
from sklearn.model_selection import train_test_split
X = X_pca
y = df['Target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size
                                                    =0.33, random_state=42)

# SVM Model Training
from sklearn.svm import SVC
from sklearn.model_selection import GridSearchCV
param_grid = {'C': [0.1, 1, 10, 100, 1000], 'gamma': [1, 0.1, 0.01,
                                                         0.001, 0.0001], 'kernel': ['rbf']}
grid = GridSearchCV(SVC(), param_grid, refit=True, cv=5)
grid.fit(X_train, y_train)
```

## 5 Insights and Conclusions:

**Key Takeaways:** As discussed in the topic motivation earlier, the goal of this project was to build a model that can accurately predict whether a tumor is benign or malignant, based on features provided in the dataset. After conducting Exploratory Data Analysis (EDA), most important features are selected to reducing the complexity of the data to improve the performance of the models. Also in SVM approach, **Principal Component Analysis (PCA)** is applied to reduce the dimensionality of the data, simplifying the problem without the loss of critical information. After conducting PCA, a **Support Vector Machine (SVM)** classifier is used, by which an accuracy of 97% is achieved, whereas its around 96% in case of Random Forest Classifier. This shows that, even with fewer features, the SVM model was able to maintain strong performance.

### Model Performance Breakdown:

**Random Forest Model:** Starting with the **Random Forest model**, here is how it performed:

$$\text{ConfusionMatrix} : \begin{bmatrix} 117 & 4 \\ 4 & 63 \end{bmatrix}$$

The classification report shows the following:

|                          | <i>Precision</i> | <i>Recall</i> | <i>F1 – Score</i> |
|--------------------------|------------------|---------------|-------------------|
| <i>Class0(Benign)</i>    | 0.97             | 0.97          | 0.97              |
| <i>Class1(Malignant)</i> | 0.94             | 0.94          | 0.94              |
| <i>Accuracy</i>          |                  |               | 0.96              |

The **Random Forest model** achieved an accuracy of 96%. For benign tumors, the precision was 0.97, meaning the model correctly predicted benign cases 97% of the time. For malignant tumors, the precision was 0.94, which is also impressive, though slightly lower than the SVM model's performance.

In terms of recall, the model was able to identify 97% of benign tumors and 94% of malignant ones. The F1-scores, that indicate the balance between precision and recall, were also solid, indicating that the model was reliable across both categories.

**PCA & SVM Model:** Moving on to the **PCA & SVM model**, here's how it performed:

$$\text{ConfusionMatrix} : \begin{bmatrix} 118 & 3 \\ 2 & 65 \end{bmatrix}$$

The classification report shows:

|                          | <i>Precision</i> | <i>Recall</i> | <i>F1 – Score</i> |
|--------------------------|------------------|---------------|-------------------|
| <i>Class0(Benign)</i>    | 0.98             | 0.98          | 0.98              |
| <i>Class1(Malignant)</i> | 0.96             | 0.97          | 0.96              |
| <i>Accuracy</i>          |                  |               | 0.97              |

The **PCA & SVM model** performed a little better, with an accuracy of 97%. The precision and recall for benign tumors resulted by this approach were both 0.98, showing that this model was very reliable in identifying benign tumors. Whereas, for malignant tumors, the precision was 0.96 and recall was 0.97, which indicates that the SVM model was also very good at identifying malignant cases. The F1-Scores for PCA+SVM for both benign and malignant cases is also more (0.98,0.96) than that of the Random Forest scores.

### Final Thoughts:

**PCA & SVM** offers **moderately strong performance** with **moderate accuracy, recall, and F1-score**, thus being a compromise when false positives and false negatives need to be minimized together. **Random Forest** is still a robust model, but its recall and F1-score have

a little less weight than those for **PCA & SVM**, implying that it might not pick up some important cases.

In conclusion, it can be said that **PCA & SVM** is more robust than **Random Forest** because it provides better overall performance, especially in terms of **accuracy**, **recall**, and **F1-score**, for this breast tumor prediction application; therefore, this will be preferred. However, the Random Forest model is a strong alternative, particularly if speed and simplicity are key priorities. Both models highlight the power of machine learning in cancer diagnosis, supporting early detection and accurate treatment planning

## Informed Reader's View:

The analysis done as part of this project shared valuable information regarding the use of machine learning algorithms such as PCA, SVM, Random Forest, in the context of their application to the breast cancer diagnosis usecase. For the informed readers, the novelty lies in the combination of SVM with PCA for the dimensionality reduction, which enhanced the computational efficiency.

This combined, novel approach resulted in robust classification and it offered useful insights about model comparison, thus opening up new possibilities for deploying such trained, evaluated methods in real-world medical applications. Our work highlights the importance of feature engineering, model selection, and the potential of machine learning models for improving the clinical outcomes.

## 6 Data Science Ethics:

### Ethical Considerations:

**Breast Cancer Wisconsin (Diagnostic) Dataset** is used for this project and we addressed the ethical concerns like **class imbalance** (more benign than malignant cases) by applying **resampling** and **class weights** to improve model fairness. Although this dataset is publicly available and anonymized, privacy remains a key concern, emphasizing the importance of data anonymization in real-world applications. The approach we followed ensured **transparency**, **fairness**, and **unbiased** analysis by:

- **Clear Documentation:** Documented all the steps, for clarity and reproducibility.
- **Model Interpretability:** We used interpretable models and assessed feature importance.
- **Bias Mitigation:** *Stratified sampling* and *class weighting* were used to address class imbalance, along with multiple performance metrics (accuracy, precision, recall, F1-score).
- **Reproducibility:** We used publicly accessible datasets, to design the analysis to be easily replicable.
- **Fairness in Evaluation:** To ensure fairness in detecting malignant tumors, we employed performance metrics that are sensitive to class imbalance.

By adhering to these principles, we ensured our analysis was ethical, transparent, and reproducible.



## 7 Additional Screenshots and Visualizations:

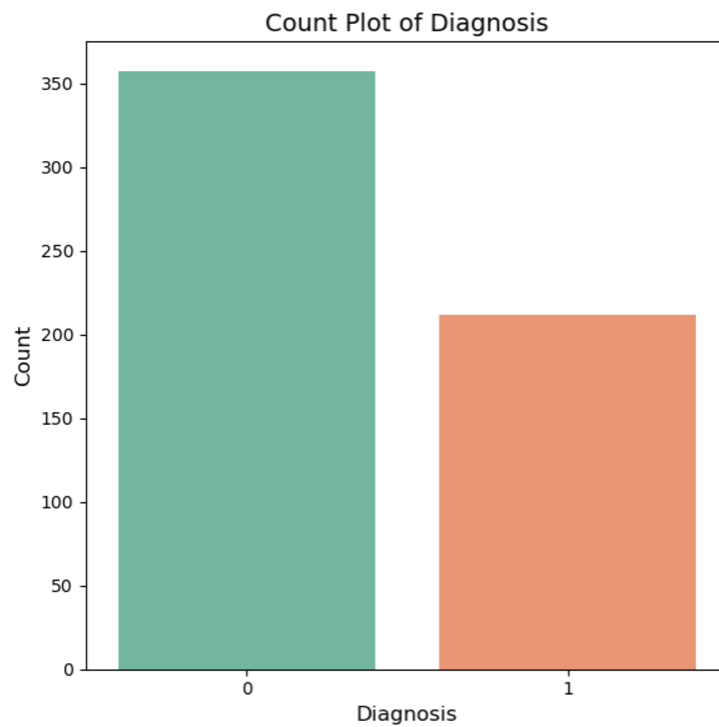


Figure 2: Count Plots for Diagnosis

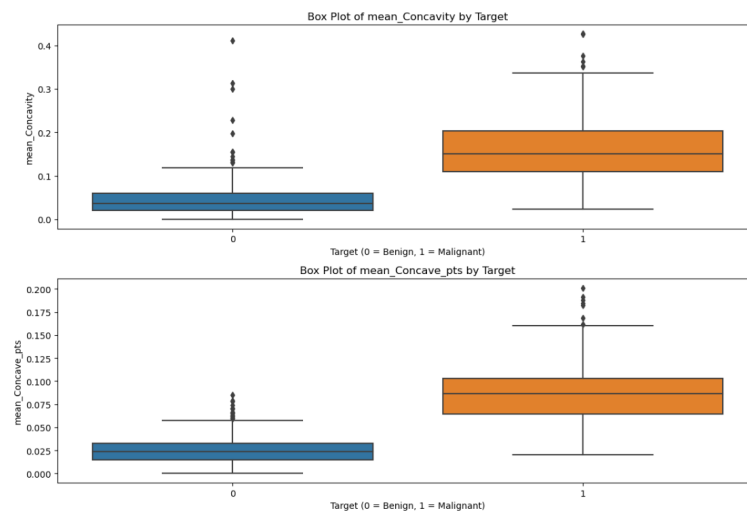


Figure 3: Box Plots for ANOVA

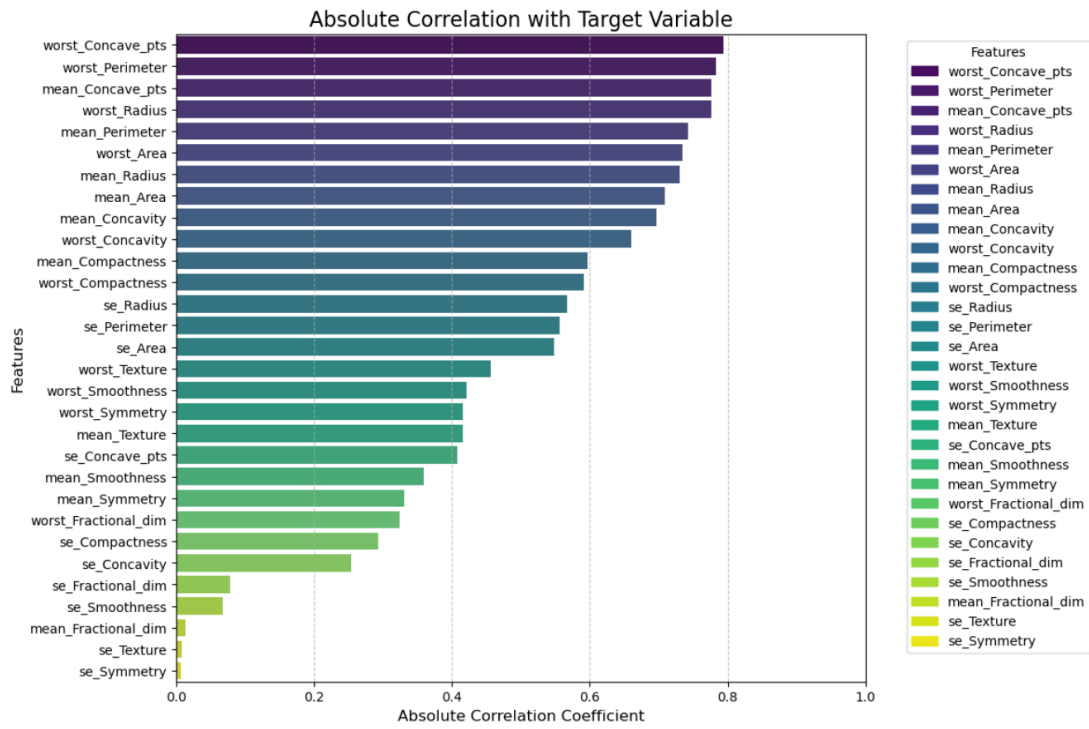


Figure 4: Correlation for Target Variable for T-Test

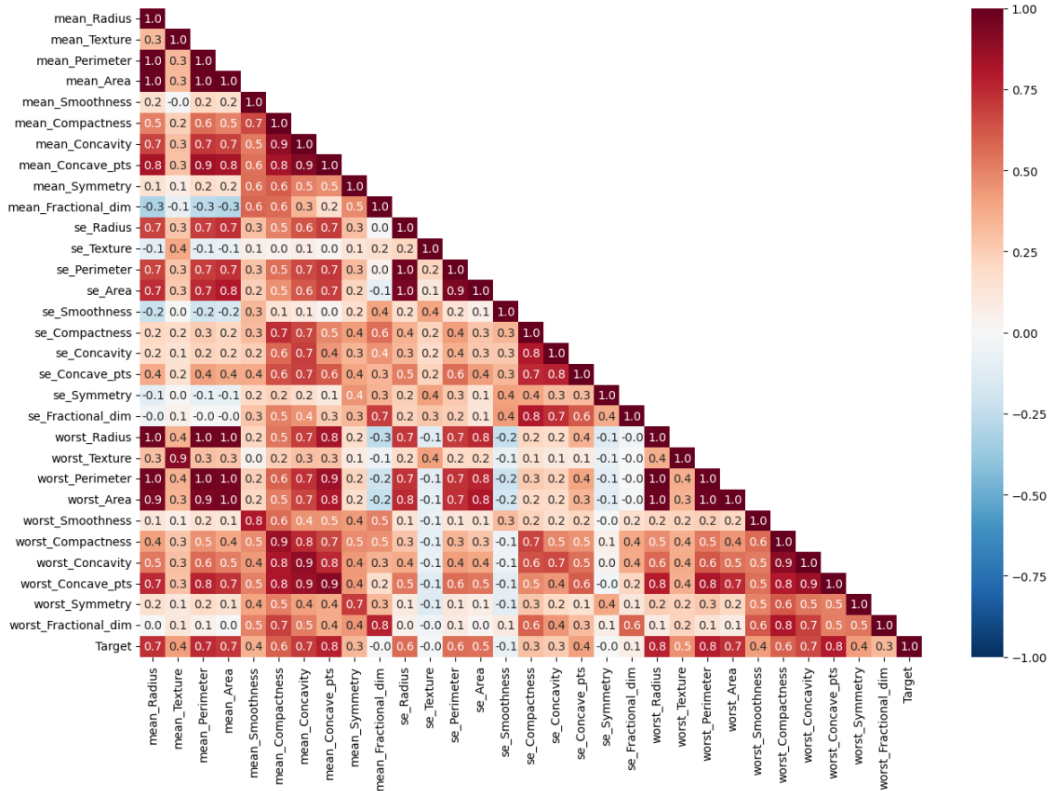


Figure 5: Correlation for Target Variable for Chi-Squared-Test

## 8 Acknowledgements:

We want to extend our sincere gratitude to Dr. Fardina Fathmiul Alam, our instructor, for her help and support during the project and the course. Her insightful comments and helpful criticism were crucial in helping us to improve our project strategy.

We are also grateful to the developers and maintainers of the Breast Cancer Wisconsin (Diagnostic) Dataset, which gave us the starting point for our study and made it possible for us to use machine learning methods to solve a practical issue.

Special thanks to the designers of open-source libraries, which include the features like Random Forest, PCA, and SVM, as well as scikit-learn, NumPy, and Pandas, which formed a crucial resource for our models implementation and assessments.

Lastly, we appreciate the wide resources and knowledge shared by the data science and machine learning community in the form of informative articles, discussions, etc., which enabled us to apply these techniques effectively.

## 9 References:

1. Wolberg, W., Mangasarian, O., Street, N., & Street, W. (1993). *Breast Cancer Wisconsin (Diagnostic)*, UCI Machine Learning Repository. Available at <https://doi.org/10.24432/C5DW2B>
2. Documentation for sklearn. *Scikit-learn: Machine Learning in Python*. Available at <https://scikit-learn.org/stable/documentation.html>
3. J. H. Friedman, T. Hastie, R. Tibshirani. (2001). *The Elements of Statistical Learning*, Springer.
4. C. Cortes, V. Vapnik. (1995). *Support-Vector Networks*. Machine Learning, 20(3), 273-297.
5. Breiman, L. (2001). *Random Forests*. Machine Learning, 45(1), 5-32.
6. Murphy, K. (2012). *Machine Learning: A Probabilistic Perspective*. MIT Press.
7. Karpathy, S., Johnson, J. (2017). *CS231n: Convolutional Neural Networks for Visual Recognition*, Stanford University. Available at <http://cs231n.stanford.edu/>
8. Verleysen, M., Francois, D. (2018). *Dimensionality Reduction: A Comparative Review*, Computational Intelligence, 21, 83-107.
9. Zhao, M., Li, X. (2020). *Improving SVM Performance in Cancer Diagnosis: Feature Selection Approaches*, Artificial Intelligence in Medicine, 45, 97-110.
10. Ross, S. (2019). *Random Forests and Their Application in Healthcare Predictive Modeling*, Healthcare Analytics, 12, 99-114.
11. Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*, Springer.
12. Zhang, H., Xie, J., & Zhao, Y. (2018). *A Comprehensive Survey on Machine Learning and Deep Learning Techniques for Breast Cancer Diagnosis*, Journal of Biomedical Informatics, 86, 45-58.
13. Kim, H., & Jeong, Y. (2020). *Improving Predictive Performance of Medical Diagnosis with Neural Networks and Feature Engineering*, Medical Image Analysis, 63, 101702.
14. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*, MIT Press.